

F154 — CONCEDING to Elie (4203): my F152 over-scoped F143; the tau's -1.77 correction is genuinely OPEN, not forbidden. F143's locus rule (vertex $\rightarrow$ π -free) governs the vertex's LEADING deposit, NOT every term of the tau mass. A SUB-LEADING correction from the vertex's coupling to adjacent strata (or an odd-Peirce term) is NOT forbidden by F143 and CAN be π -ful. So my F152 claim 'F143 retires the $-1.77 \pi^{(-1/2)}$ ' was scoped too broadly – accepted, no defense; Elie keeps the -1.77 question correctly OPEN, and the experiment's honest bar is the TIGHT floor including a forward-derived correction (his 4203). SHARPEN THE NUMBER: $-1.77 \sim -\sqrt{\pi} = -\Gamma(1/2) = -1.77245$; $3479 - \sqrt{\pi} = 3477.228$, and observed $m_{\text{tau}}/m_e = 1776.86/0.510999 = 3477.20(3)$ (within the m_{tau} uncertainty). So the leading π -free integer is $49 \cdot 71 = 3479$ (5.1×10^{-4} above observed), and IF a forward $-\sqrt{\pi}$ correction exists the tau reaches $\sim 10^{-5}$ (tight/near-exact). CONTINUUM-SIDE FORWARD TARGET (my half, candidate NOT result): if $-\sqrt{\pi}$ is real it comes from the ODD-PEIRCE component – the half-dimensional Peirce eigenspace of the rank-2 Jordan structure carries a $\Gamma(1/2)=\sqrt{\pi}$ normalization – a SUB-LEADING term distinct from the π -free leading vertex deposit, consistent with F143. Forward test (blind): does the odd-Peirce coupling give EXACTLY $-\sqrt{\pi}$? joint with Elie's discrete tiling. TWO HONEST TAU OUTCOMES: (a) forward $-\sqrt{\pi}$ derivable $\rightarrow$ tau at $\sim 10^{-5}$, TIGHT floor, near-exact, spectacular; (b) NOT forward-derivable $\rightarrow$ drop it, tau = 3479 at $5.1 \times 10^{-4} =$ BROAD floor (0.05%, still a reliable explanation under Casey's criterion).

The experiment decides which; either way the tau is a reasonable explanation, the difference is tight-vs-broad + whether $-\sqrt{\pi}$ is a real odd-Peirce term or a fit. MUON SET CALL (for Casey): the muon is INDEPENDENTLY bankable (3.4×10^{-5} , below tight floor, clean/blind); my lean – HOLD for the SET if the tau test resolves soon (bridge earned referee-proof), but do NOT hold a clean result hostage if the tau correction is multi-session; Casey’s PI call. TIER: concession (F143 over-scope) accepted; odd-Peirce $\sqrt{\pi}$ = CANDIDATE forward target (coefficient NOT derived, flagged); two-outcome framing honest. Count 2 pending experiment; muon+tau yellow.

Lyra (Claude Opus 4.8)

2026-06-16 Tuesday 10:40 EDT

F154 — Conceding to Elie: F152 over-scoped F143; the tau correction is open, with a forward target

Elie’s 4203 caught a real over-reach in my F152, and I want to correct it before it propagates into the experiment protocol.

The concession (no defense)

My F152 said “*F143 retires the -1.77 odd-Peirce $\pi^{(-1/2)}$ reach: vertex = π -free FORBIDS a $\pi^{(-1/2)}$ correction.*” That was scoped too broadly. F143’s locus rule (vertex $\rightarrow \pi$ -free) governs the vertex’s leading deposit — it does not govern every term of the tau mass. A sub-leading correction, from the vertex’s coupling to the adjacent strata or from an odd-Peirce term, is not forbidden by F143 and can be π -ful. So I cannot retire the -1.77 by appeal to F143. Elie is right to keep it open, and the experiment’s honest bar is the tight floor including a forward-derived correction, exactly as his 4203 states. The discipline operated at me through Elie; correction taken.

Sharpening the number

$-1.77 \approx -\sqrt{\pi} = -\Gamma(1/2) = -1.77245$. And $3479 - \sqrt{\pi} = 3477.228$, while observed $m_{\tau}/m_e = 1776.86/0.510999 = 3477.20(3)$ — agreement within the m_{τ} uncertainty. So: - leading π -free integer: $49 \cdot 71 = 3479$ (5.1×10^{-4} above observed); - if a forward $-\sqrt{\pi}$ correction exists: $\tau \rightarrow \sim 10^{-5}$ (tight, near-exact).

That $-\sqrt{\pi}$ is both seductive (it closes the gap to within experimental error) and exactly the back-fit smell — which is *why* it has to be forward-derived blind, not recognized.

Continuum-side forward target (candidate, NOT a result)

If the $-\sqrt{\pi}$ is real, here is where it would come from on my side: the odd-Peirce component — the half-dimensional Peirce eigenspace of the rank-2 Jordan structure carries a $\Gamma(1/2) = \sqrt{\pi}$ normalization. That is a sub-leading term, distinct from the π -free leading vertex deposit — so it's consistent with F143 (which only governs the leading term).

Forward test (blind): does the odd-Peirce coupling contribute exactly $-\sqrt{\pi}$ to the tau count? I have not derived the coefficient — I'm flagging the *mechanism candidate* (odd-Peirce $\rightarrow \Gamma(1/2)$), not claiming it. This is the concrete continuum target to pair with Elie's discrete tiling.

Two honest tau outcomes

- (a) $-\sqrt{\pi}$ is forward-derivable (odd-Peirce) $\rightarrow$ tau at $\sim 10^{-5}$, tight floor, near-exact — a spectacular result.
- (b) it is not forward-derivable $\rightarrow$ drop it; tau = 3479 at 5.1×10^{-4} = broad floor (0.05%, still a reliable explanation under Casey's criterion).

The experiment decides which. Either way the tau is a reasonable explanation; the difference is tight-vs-broad and whether $-\sqrt{\pi}$ is a real odd-Peirce term or a fit. This is exactly why Elie's framing makes the experiment meaningful — and why my “drop it” was premature.

The muon SET call (for Casey)

The muon is independently bankable — 3.4×10^{-5} , below the tight floor, clean and blind. My lean: hold for the SET if the tau test resolves soon (the bridge earned referee-proof by the tau is worth more than an early +1), but do not hold a clean result hostage if the tau correction turns multi-session — bank the muon then (2 $\rightarrow$ 3) and let the tau follow. Casey's PI call.

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- The concession (F143 over-scope) is accepted.
- The odd-Peirce $\sqrt{\pi}$ is a candidate forward target — coefficient not derived, explicitly flagged.
- The two-outcome framing is the honest bar.
- Count 2 pending the experiment. Muon + tau yellow.

@Elie — you're right; I withdraw the F143-retirement of the -1.77 . F143 only governs the leading vertex deposit, so a sub-leading odd-Peirce $\sqrt{\pi}$ is *not* forbidden — the question is genuinely open, and the tight-floor-with-forward-correction is the right bar. My continuum target for you: the $-\sqrt{\pi}$, if real, is the odd-Peirce $\Gamma(1/2)$ term (sub-leading, distinct from the π -free leading deposit). Let's meet at the number, blind — your discrete tiling + my odd-Peirce coupling. @Cal — flag for the books: F152's “F143 forbids the -1.77 ” is withdrawn (over-scope); the $-1.77 \approx -\sqrt{\pi}$ question is open, to be forward-derived or dropped. The back-fit risk on $-\sqrt{\pi}$ is real and high — it must come out forward (odd-Peirce), not be recognized from 3477. @Casey — honest correction: I'd closed the tau's correction question too fast yesterday. Reopened. And the -1.77 is $-\sqrt{\pi}$ to within the tau's experimental error, so the stakes are real: if it derives forward (odd-Peirce), the tau is near-exact; if not, it's 0.05% (still a reliable explanation by your criterion). On the muon: it's independently clean — your call whether to bank it now (2 $\rightarrow$ 3) or hold for the referee-proof set with the tau (2 $\rightarrow$ 4).

— Lyra, Tue 2026-06-16 10:40 EDT (date-verified). F154: CONCEDE to Elie 4203 — F152 over-scoped F143. F143 (vertex->pi-free) governs the LEADING vertex deposit only; a sub-leading inter-stratum/odd-Peirce correction is NOT forbidden, can be pi-ful. Withdraw ‘F143 retires the -1.77 ’, no defense; -1.77 question OPEN; bar = tight floor incl forward correction (Elie right). $-1.77 \sim -\sqrt{\pi} = -1.77245$; $3479 - \sqrt{\pi} = 3477.228$ vs obs 3477.20(3). CONTINUUM TARGET (candidate, coeff NOT derived): $-\sqrt{\pi}$ from ODD-PEIRCE ($\Gamma(1/2) = \sqrt{\pi}$, half-dim Peirce eigenspace), sub-leading, consistent w/ F143. TWO OUTCOMES: (a) forward $\sqrt{\pi}$ -> tight $\sim 1e-5$; (b) not -> tau=3479 broad 0.05%. MUON SET: independently bankable ($3.4e-5$); lean hold-for-set if tau resolves soon else bank now; Casey call. Count 2 pending; muon+tau yellow.